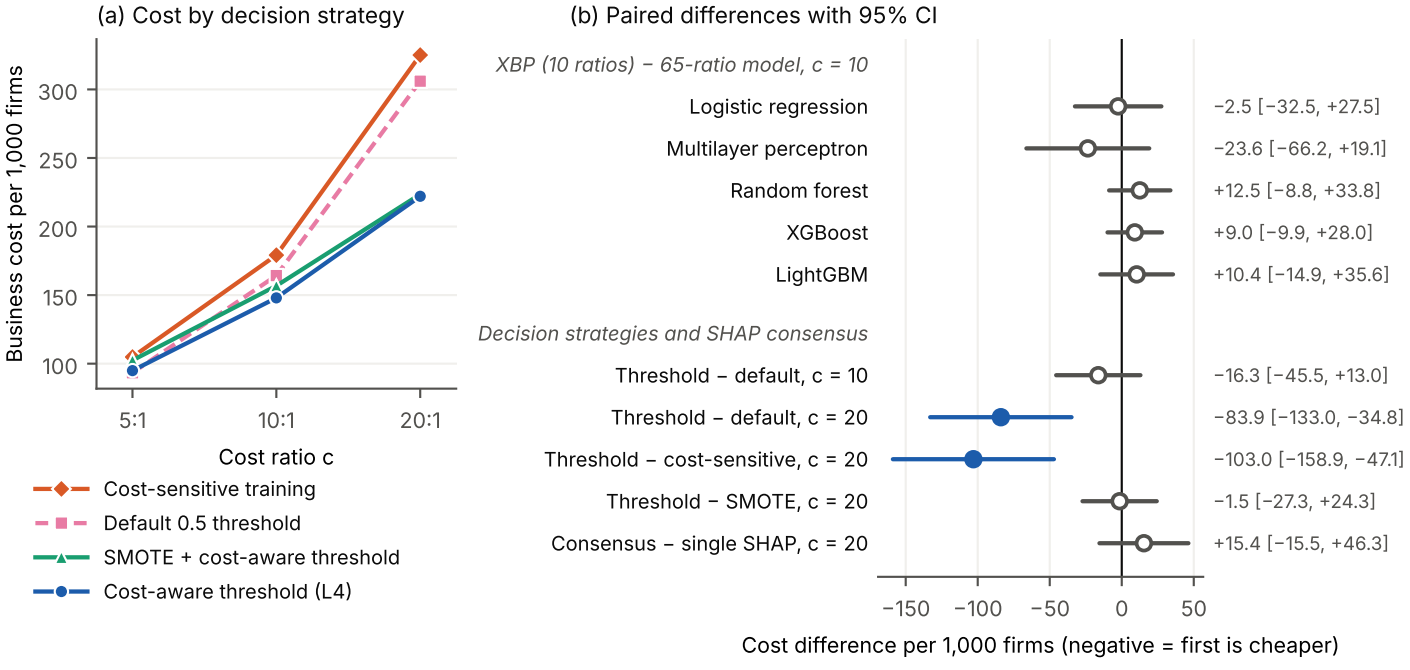
(a) Cost by decision strategy
Business cost per 1,000 firms
300
250
200
150
100
5:1
10:1
20:1
Cost ratio c
Cost-sensitive training
Default 0.5 threshold
SMOTE + cost-aware threshold
Cost-aware threshold (L4)
(b) Paired differences with 95% CI
XBP (10 ratios) − 65-ratio model, c = 10
Logistic regression
Multilayer perceptron
Random forest
XGBoost
LightGBM
Decision strategies and SHAP consensus
Threshold − default, c = 10
Threshold − default, c = 20
Threshold − cost-sensitive, c = 20
Threshold − SMOTE, c = 20
Consensus − single SHAP, c = 20
−2.5 [−32.5, +27.5]
−23.6 [−66.2, +19.1]
+12.5 [−8.8, +33.8]
+9.0 [−9.9, +28.0]
+10.4 [−14.9, +35.6]
−16.3 [−45.5, +13.0]
−83.9 [−133.0, −34.8]
−103.0 [−158.9, −47.1]
−1.5 [−27.3, +24.3]
+15.4 [−15.5, +46.3]
−150
−100
−50
0
50
Cost difference per 1,000 firms (negative = first is cheaper)